Morphological Segmentation to Improve Crosslingual Word Embeddings for Low Resource Languages

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Crosslingual word embeddings developed from multiple parallel corpora help in understanding the relationships between languages and improving the prediction quality of machine translation. However, in low resource languages with complex and agglutinative morphologies, inducing good-quality crosslingual embeddings becomes challenging due to the problem of complex morphological forms and rare words. This is true even for languages that share common linguistic structure. In our work, we have shown that performing a simple morphological segmentation upon the corpora prior to the generation of crosslingual word embeddings for both roots and suffixes greatly improves the prediction quality and captures semantic similarities more effectively. To exhibit this, we have chosen two related languages: Telugu and Kannada of the Dravidian language family. We have also tested our method upon a widely spoken North Indian language, Hindi, belonging to the Indo-European language family, and have observed encouraging results.

CCS Concepts: • **Computing methodologies** → *Machine translation*; *Phonology/morphology*;

Additional Key Words and Phrases: Word embeddings, word2vec, crosslingual embeddings, machine translation, morphology, morphologically rich languages, bilingual embeddings, supervised learning, linear transformation

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1 INTRODUCTION

Neural network methods for natural language processing (NLP) tasks such as text classification, summarization, sentiment analysis, and machine translation are gaining wide importance in the present scenario. Neural machine translation (NMT) systems [Bahdanau et al. 2014; Sutskever et al. 2014; Cho et al. 2014] have shown promising results in producing fluent and natural translation outputs [Wu et al. 2016]. All NMT systems use some variant of the neural probabilistic language model [Bengio et al. 2003], which converts a word to its vector representation (known as word embeddings) and thereby captures the context information. Mikolov's model for generating embeddings, namely word2vec [Mikolov et al. 2013a; Mikolov et al. 2013c] has become highly

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popular due to its speed and flexibility, also leading to an increased interest in generating bilingual and multilingual word embeddings.

Bilingual word embeddings have shown to improve crosslingual tasks such as sentiment analysis and document classification [Gouws and Søgaard 2015]. Bilingual word embeddings effectively capture the semantic relationships within a single language and across two languages [Mikolov et al. 2013d]. They also have the ability to identify morphologically related words [Soricut and Och 2015]. This makes them an ideal feature for machine translation [Zou et al. 2013].

In their later work, Mikolov et al. [2013c] utilized their word2vec model to generate crosslingual embeddings by exploiting similarities among languages. Several enhancements for generating word embeddings have been proposed henceforth [Faruqui et al. 2014; Luong et al. 2015b; Xing et al. 2015; Lu et al. 2015].

However, although these approaches have predominantly dealt with high resource languages and huge corpora, extending them for generating crosslingual predictions across low resource languages is problematic. In addition, within the limited corpora available, the embeddings tend to be very sparse due to rich, agglutinative morphology for some languages. There is a large number of morphological variants for each root word, most of which rarely occur in small training sets. This is especially true for Indian languages. Although the language pairs are similar, crosslingual tasks like machine translation often yield poor results due to these issues. We have observed that crosslingual embeddings that are extracted at word level fail to capture the morphological and semantic relatedness between the words and mostly predict a large number of unrelated words as candidate translations.

In our work, we propose a solution that extends the work in Mikolov et al. [2013c] to develop embeddings at the morpheme level rather than word level to improve the crosslingual prediction on three Indian languages: Telugu, Kannada, and Hindi. By performing a simple morphological segmentation prior to the generation of embeddings, and generating embeddings separately for suffixes and roots, we were able to achieve better similarity scores and accuracies for target language predictions and effectively capture crosslingual semantic similarities, thus resulting in better prediction quality when compared to the word-level embeddings. We will use the word *morph* henceforth to refer to both lemma and suffix as well.

The rest of the article is organized as follows. In Section 2, we enumerate the challenges of complex morphology in the chosen languages and then proceed to explain the methods we used. In Section 3, we present our evaluation methodology. Section 4 presents the results obtained. Section 5 is dedicated to discussing these results. Section 6 describes the related work with regard to our problem. Section 7 concludes the article and suggests directions for future work.

2 METHODOLOGY

2.1 Challenges Posed by Dravidian Languages

Two of the languages of our choice, Kannada and Telugu, belong to the Dravidian language family, with 50 million and 75 million speakers, respectively, spread across India and various parts of the world. The other language of choice is Hindi, widely spoken by 551 million speakers across India, belonging to the Indo-European language family. Telugu and Kannada are highly agglutinative compared to Hindi. Despite a large number of speakers, these languages remain scarce in terms of language processing resources. Machine translation among Dravidian languages themselves and also other Indian languages turns out to be challenging due to their rich morphology and agglutinative nature.

Table 1. Some Morphological Variants of the Kannada Verb *māḍu*

Verb Form	Meaning	Inflection
māḍi-de-(nu)	I did	1st person singular
māḍ-utt-ēne	I do/I will do	1st person singular
māḍi-de-vu	We did	1st person plural
māḍ-utt-ēve	We do/We will do	1st person plural
māḍ-id-e	You did	2nd person singular
māḍ-id-a(nu)	He did	3rd person singular (m)
māḍ-id-aḷu	She did	3rd person singular (f)
māḍ-i-tu	It did	3rd person singular (n)

In agglutination, suffixes are added to the root words to derive different meanings. We have identified that there are three major sources of agglutination in Dravidian languages:

- *Vibhakti*: Case markers that get attached to the nominal
- *Sandhi*: Morphophonemic changes when two words are fused together
- Orthographic interactions.

In all Dravidian languages, the nominals (nouns, pronouns, numerals, and adverbs of time and space) are inflected for case [Krishnamurti 2003]. These cases are known as *vibhakti*s. For example, the dative case of a noun is formed by adding *-kke* and *-ge* in Kannada. The accusative case is formed by the suffix *-annu*. Similar construction of the dative case noun is done by adding *-ki* and *-ku* in Telugu. The accusative case adds *-ni* and *-nu* to the nominal. Nouns and pronouns also inflect for gender and number. Adverbs and numerals become adjectives via suffixation [Krishnamurthi 2003].

The morphology of verbs is even more complex. A verb form is constructed as (Root + (transitive) + (causative)) + Tense + Gender-Number-Person [Krishnamurthi 2003].

For the Kannada root word *māḍu* (meaning "to do") in Kannada, *māḍi-de-nu* represents 1st person singular past tense. *māḍu-tte-ne* represents 1st person singular non–past tense. More examples of the morphological variations can be seen in Table 1. Telugu also exhibits similar compounding in verbs. For example, the root word *wanḍu* (meaning "to cook") takes several forms, such as *wanḍaaḍu* (3rd person male singular) and *wanḍaamu* (1st person plural inclusive). There are about 300 commonly occurring forms for each verb root. There are also several auxiliary verbs observed in Dravidian languages that add meaning when attached to the main verb but act differently when used separately. For example, *biḍu*, a verb in Kannada, literally meaning "to leave" is used to emphasize the completion of an action when used as an auxiliary [Krishnamurthi 2003].

Sandhi is the morphophonemic change occurring when two words are fused together. *Sandhi* results in addition, omission, or modification of a syllable. For example,

- mā + amma = maayamma

(The *ya* syllable gets added in the middle due to the typical *sandhi* rules.)

- maha + īṣwara = mahēṣwara

(The addition of new vowel *ē* due to *sandhi* rules.).

Other orthographic interactions are typical to each language and occur as a result of syllable overlap in colloquial language. For example, a Telugu phrase, *nenu ippudu* (meaning "me now") can also be written as a single word, *nenippudu*, due to such syllable overlap. Word embeddings struggle to identify these as meaning the same.

Addition of participles, negations, and clitics (emphatic or interrogative) make the morphology quite rich, and the number of words in the vocabulary is extremely huge to handle for limited size corpora. Handling all of these forms using consistent rules and a finite state machine is possible but proves expensive. Mapping these phenomena among languages is even more challenging.

Further, all three languages exhibit word compounding due to the direct influence of Sanskrit. Long words like *niyamanibandhana* (meaning "rules and regulations," composed of two words *niyama* and *nibandhana*, both derived from Sanskrit) are common enough in formal speech and writing in most Indian languages.

Despite having similar word order in sentences and an almost parallel morphological structure, translation among Dravidian languages is not easy. Translation to and from the Dravidian family to other Indian languages is not accurate either. This is shown to be because of the aforementioned issues of complex morphology [Kunchukuttan et al. 2014].

2.2 Our Approach

The basic word2vec implementations [Mikolov et al. 2013a, Mikolov et al. 2013c] treat the whole word as a single unit. In small-sized corpora, multiple inflections of same lemma/root have very low frequencies, leading to poor-quality embeddings. In such cases, word embeddings are inefficient in predicting the morphological variants and semantically similar words. When used as features for the machine translation task, they tend to predict totally unrelated words as target translations, thus resulting in translations that lack both accuracy and fluency. Thus, to tackle this problem and make the generation of crosslingual embeddings more morphology aware, we suggest performing a morphological segmentation upon the corpus prior to training the embeddings. In addition, two of the languages of our choice are both agglutinative and have a rich inflective morphology. Each morph that is attached to the lemma adds a part of the meaning to the word. This means that a lot of information needed for performing part-of-speech (POS) tagging of the language is implicitly encoded within the word forms. This is an added advantage of making the word embeddings more morphologically aware. We explain this in more detail in the following sections.

In short, our approach can be summarized into the following steps:

- Identify and segment suffixes
- Create bilingual dictionaries
- Train translation matrix using bilingual dictionaries
- Generate translations.

The generation of translations for word models follows all of the preceding steps except for the first one.

2.3 Selection of Corpus

As a preliminary study, we tested our approach on a readily available parallel text, the Holy Bible in Telugu and Kannada (http://christos-c.com/bible/). These corpora are mostly sentence aligned—that is, the sentences carry the same meaning although the word order is divergent. We took the first 5,000 sentences of each Bible for our work. The number of unique tokens was about 7,000 in both the languages. We obtained promising initial results with the CBOW variant of embeddings on the morph-segmented corpus. This encouraged us to repeat our experiments on larger corpora.

Currently, Technology Development for Indian Languages (TDIL),[1] an Indian government organization, makes parallel corpora freely available across multiple Indian languages. We compiled our corpus from Hindi-Telugu and Hindi-Kannada parallel corpora, comprised of sentences in agriculture, entertainment, history, and tourism domains. We also observed that the vocabulary had a huge variety of morphological variants ranging across various domains, making it more ideal to our task when compared to the Bible, which falls short in terms of such diversity and tended to overfit due to the lack of variety. The Telugu corpus contains 31k sentences with about 67k unique tokens. Similarly, the Kannada corpus has 34k sentences with 67k tokens. The Hindi corpus has 40k sentences with only 39k unique tokens. Since Hindi is nonagglutinative and exhibits fewer inflections per lemma, it can be observed that the vocabulary explosion problem is not very pronounced in this case.

2.4 Morphological Segmentation

As mentioned previously, there is an enormous number of forms possible for each root word in Telugu and Kannada due to their agglutinative nature. The best way to obtain an accurate segmentation would be to hand code the rules for all cases, *sandhi*s, and orthographic interactions. However, as this approach demands highly specific rule writing for each language, we decided to go for a language-independent segmentation model, Morfessor [Creutz and Lagus 2005]. This approach is statistically driven and makes use of the frequency of the occurrence of suffixes.

When we tried to segment the Bible vocabulary using Morfessor, even though the algorithm was able to generate correct splits for simple morphs ranging from one to two characters, we detected that several complex morphs were improperly segmented by the algorithm. Hence, we had to perform a manual postprocessing by editing these incorrect morph segmentations. Although this does not affect results in smaller corpora, it might not be scalable to large corpora.

The structure of Dravidian languages is such that certain suffixes are specific to words of a particular POS tag. For example *vibhakti*s or case markers only get attached to nouns and pronouns. Similarly, gender- and tense-specific morphs only occur in verbs. Therefore, we adopted a POS tag–specific suffix segmentation approach for arriving at fairly consistent splits. We performed a POS tagging of the corpora by Reddy and Sharoff [2011], a TnT-based POS tagger available for Kannada, Telugu, and Hindi.[2] Using this, we grouped our vocabularies into various POS tags. We then performed an unsupervised morphological analysis on each group using Morfessor. From Morfessor's results, we identified approximately 450 common suffixes (both simple and compound) in each language, of which around 300 are specific to verb forms alone. Thus, we created a segmentation dictionary that maps a word into its constituent morphs. This has streamlined the morpheme segmentation to a large extent, minimizing the human effort in postprocessing.

Following this approach, we created our morph corpora, containing 37k Telugu morphs, 34k Kannada morphs, and 37k Hindi morphs.

*Sandhi*s, compound words, and orthographic variations were not explicitly handled.

2.5 Bilingual Dictionary

The objective of learning crosslingual embeddings is to project vectors representing words of various languages into a shared embedding space that captures crosslingual similarities, thereby aiding crosslingual tasks. The method used for such projection in Mikolov et al. [2013c] is a linear transformation from one vector space to another using a translation matrix. This translation matrix is trained using a gold standard dictionary, which is compiled using human speakers' inputs. The

[1] http://tdil.meity.gov.in/.
[2] http://sivareddy.in/downloads.

training objective in Mikolov et al. [2013c] was that the words that are translations of each other are geometrically closer in a common vector space. The translation matrix is generated as follows:

$$\min_{W} \sum_{i=1}^{n} ||Wx_i - z_i||^2, \tag{1}$$

where W is the required translation matrix, x is the source language embedding, and z is the target language embedding. W is learned by minimizing the squared Euclidean distance between Wx and z.

This trained translation matrix W is further used to make translation predictions for words that are outside the training vocabulary.

In our case, we used a 560-word-pair bilingual dictionary for the small Bible corpus. For the larger TDIL corpora, we created a 1,000-tuple dictionary for three-way parallel translations among Telugu, Kannada, and Hindi entities. In addition to translations for single words and two-word phrases like that in Mikolov et al. [2013c], we also included multiword phrases and sentence fragments that carry the same meaning across the three languages. The embeddings x and z in such cases are taken to be the average of all word embeddings in the phrase/fragment. The same training dictionary was adopted for morph models as well, where each language entry was segmented into corresponding morphs using suffix segmentation as described earlier. The same averaging strategy was followed for morphs.

2.6 Generation of Word Embeddings

We trained both 300-dimensional embeddings on both word and morph variants of the TDIL corpora. The window size used was 5 for the word model and 11 for morph models to include an equivalent context in both. The negative sampling parameter was 5, and the models were trained for 100 iterations.

3 EVALUATIONS

It is desirable to build crosslingual embeddings that also capture regularities in a monolingual scenario. To examine how well our strategy of segmentation helps monolingual tasks for the agglutinative languages, we performed both intrinsic and extrinsic evaluations of monolingual word embeddings in Telugu and Kannada. For intrinsic evaluation, we chose a word similarity task. The top 10 similar words of the embedding model for a test set of 100 words were ranked by human speakers on a scale of 3 to 0, from most similar to least similar. Spearman's rho was used as the correlation measure for this purpose. In the case of morph embedding models, we excluded suffixes from the similarity evaluation test set.

For extrinsic evaluation of monolingual embeddings, we set up a task of document classification. We collected 1,200 documents in Telugu and 1,400 documents in Kannada ranging across five topics. The document embeddings were calculated as an average of the token embeddings in both word and morph cases, weighted by the TF-IDF frequencies.

Finally, for crosslingual evaluations, we predicted translations from Kannada to Telugu and Hindi to Telugu. We evaluated both the top 1 and top 5 accuracies of the predicted translations. We also evaluated Spearman's rho for human rankings on a 5-point Likert scale, similar to that used in SemEval tasks. We asked speakers to rank the candidate translations from 5 to 1, with 5 representing exact translations or close synonyms, 4 for lexical similarity (as in *do, doing*, and *done*), 3 for semantic similarity (as in *bird, animal*, and *fish*), 2 for words that might co-occur in certain contexts (e.g., *bird* and *wing*), and 1 for totally unrelated words. These ranks were later correlated with the generated cosine similarities to derive Spearman's rho coefficient values.

Table 2. Monolingual Intrinsic and Extrinsic Evaluation Results

Model	Spearman ρ (Intrinsic)	Document Classification Accuracy
Kannada word	0.13	85.5%
Telugu word	0.29	83.4%
Kannada morph	**0.32**	**95%**
Telugu morph	**0.32**	**98.3%**

4 RESULTS

By testing our method upon the Bible parallel text, we observed that the best top 1 and top 5 crosslingual accuracies were obtained by the CBOW model trained upon the morpheme-segmented corpus. This is because our intuition in inducing crosslingual embeddings is to take advantage of similar context words across multiple languages. CBOW variants perform the best for this purpose, as they average over all context word vectors to predict the target word. However, skipgram models predict the set of context words given a target word, but word order divergence and free word order in the languages of our choice can result in diverse sets of contexts for the same target word. This proved problematic for the crosslingual transfer of embeddings in the skip-gram case, resulting in very low prediction accuracies and a large number of dissimilar words in the predictions. Therefore, we opted to evaluate and compare only the CBOW variants for all of our tasks.

In the monolingual intrinsic evaluation, we retrieved the top 10 similar words for both the word and morph models in Telugu and Kannada, ordered from closest to farthest using the metric of cosine distance. The correlations of cosine distance with human speaker ranks on a scale of 3 to 0 were used to derive Spearman's rho in each case, with 3 as the rank for both lexically similar words and synonymous words, 2 for semantically related words, 1 for words that might occur in same context, and 0 for unrelated words. The number of words in each test set was 100.

Next, we evaluated our monolingual embeddings in an extrinsic evaluation task of document classification. We collected 1,200 documents in Telugu and 1,400 documents in Kannada from various web portal newspapers. We performed our suffix segmentation to create the corresponding morpheme-segmented documents. These documents were each of about 300 words in length and ranged across five different topics: movies, sports, health, travel, and food. For each labeled document, document embedding was calculated by averaging the embeddings of tokens (either word or morph) using the TF-IDF weights of the tokens. Finally, a five-way classification was performed using support vector machine with a linear kernel. The training-test split was 80–20 in each case. Both the intrinsic and extrinsic evaluation results are presented in Table 2.

There is a similarity dataset for Indian languages proposed in Akhtar et al. [2017], containing similarity scores for seven Indian languages, not including Kannada. The publicly available Telugu test set contains about 70 distinct words. We present some of our results compared with the results from the preceding test set in Table 3. Since multiple correct answers are possible and some of the words have not been covered in our own corpus, we present the predicted words and their meanings along with the similarity scores in Table 3. The words are transliterated to Latin characters with meaning mentioned in parentheses. The similarity scores are indicated using numbers. It is evident that the predictions generated by our model are quite reasonable in terms of monolingual similarity and relatedness.

For the crosslingual case, we have three evaluation measures across two different test sets. It is presumed that Dravidian languages benefit from morpheme segmentation prior to crosslingual embedding generation due to their similar morphological structure. We wanted to try the same

Table 3. Comparison with Similarity Dataset Results for Telugu Words

Word	Results from Akhtar et al. [2017]	Results from Our Model
pustaka (book)	*kāgitaṁ*—8 (paper)	*kavi*—7 (poet)
kampyūṭar (computer)	*kībōrḍu*—5 (keyboard)	*grāphiks*—7 (graphics)
vimānaṁ (airplane)	*kāru*—5 (car)	*digi*—5.3 (landed)
mīḍiyā (media)	*rēḍiyō*—6.5 (radio)	*priṇṭu*—6.1 (print)
ṭennis (tennis)	*rākeṭ*—8 (racket)	*krīḍakāruḍu*—7.5 (player)
caṭṭaṁ (law)	*n'yāyavādi*—8 (lawyer)	*savaraṇa*—6.5 (amendment)
Āhāra (of or relating to food)	*paṇḍu*—6 (fruit)	*dhān'yāla*—7 (grains)

Table 4. Crosslingual Evaluation Results

Model	Top 1	Top 5	Spearman Rho
Kan2Tel word	7.72%	11.36%	0.101
Kan2Tel morph	**21.8%**	**31.8%**	**0.210**
Hin2Tel word	Negligibile	Neglibile	~0
Hin2Tel morph	**4.7%**	**15.4%**	**0.15**
Kan2Tel word (f)	2%	7%	**0.11**
Kan2Tel morph (f)	5%	10%	**0.09**

approach on Hindi, which has an entirely different morphological structure and word order as well. We attempted to predict translations from Kannada to Telugu (Kan2Tel) and Hindi to Telugu (Hin2Tel). In each case, we calculated the top 1 and top 5 accuracies along with the Spearman rho coefficient. The test sets comprised 200 words in both cases. This measure was chosen because even when the top 1 and top 5 accuracies were poor for certain words, it was observed that there were multiple related words retrieved as candidate translations. Higher word similarity scores (both lexical and semantic) will improve the fluency and understandability of translations. For example, when the gold standard target translation is "I am leaving," a predicted translation such as "I am left" makes more sense to the reader than when *leaving* is predicted as some other unrelated word like *phone* or *red.* Since *left* is a verb form related to *leaving,* it carries more sense in a translation than any other semantically unrelated word. This is analogous to human speakers who are not fluent in the language but try to communicate the sense from their limited and often erroneous vocabulary, where they compensate the lack of knowledge of the specific word form by making use of some known closely related word form. Hence, we wanted to identify which model had the best similarity score, particularly in low resource conditions.

Along with the chosen word2vec baseline, we also computed fasttext embeddings [Bojanowski et al. 2016] for both word and morph models in Kannada to Telugu translations. Fasttext embeddings are designed to handle long and rare words by using character n-grams instead of word-level embeddings. Therefore, we aimed to test how these work for the morphologically complex pair—that is, Kannada and Telugu in our case. These results are illustrated in Table 4 (The best results are presented in bold). The results for fasttext embeddings are indicated by (f) in the last two rows.

While evaluating morph models, we predicted the translations for each component of the segmentation (i.e., for the lemma and suffix(es) separately). Suffixes have shown extremely high accuracy amounting to almost 90% due to their high frequency. Hence, while evaluating, we excluded them from our test set. We discuss these results in the following section.

5 DISCUSSION

5.1 Monolingual Evaluations

The goal of our experiment was to encourage the standard word embeddings to be more aware of the word morphology for complex languages and thereby be proven useful for crosslingual tasks even on small corpora. While improving crosslingual similarity, we also wanted to understand how well morpheme segmentation helps in monolingual scenarios. It is evident from the Spearman's rho values in Table 2 that morphological segmentation definitely improves the quality of monolingual embeddings. In Telugu and Kannada, words of the same lemma tend to be farther apart, as each form is considered a separate word by the embedding model. When these complex forms were reduced to their corresponding lemma + suffix(es) form, lemmas of closer meanings and related words seemed to appear in the predictions. This was observed to be more valid for the case of verbs in both Telugu and Kannada. For example, a Kannada verb ಕೊಡಿ (*koḍi*, meaning "to give") produced ಕಳಿ (*kaḷi,* "send") and ಕಳುಹಿ (*kaḷuhi*, "sent") as the closest words when queried on the morph model. Comparing these to ಗುರುದ್ವಾರದಿಂದ (*gurudvāradinda*, "from the Gurudwara") and ರಾಣಿಯಾಗಿ (*rāṇiyāgi*, "as a queen"), which were predicted by the word model, we understand that morpheme segmentation improved the implicit semantic knowledge encoded within the embeddings. Similar cases were observed in Telugu as well: చేస్తు (*cēstu*, a verb fragment indicating the meaning "to do") has ఇస్తు (*istu,*"to give"), తెస్తు (*testu*, "to bring"), and ఉంటు (*uṇṭu*, "to be") as its closest neighbors in the morph embedding model, whereas word embedding models predict the entirely unrelated బంగాళాదుంపల (*baṅgāḷādumpala*, "of potatoes") and సభ్యులుకూడా (*sabhyulukūḍā*, "members too") as the neighbors. We can conclude from these results that we have stepped toward improving monolingual similarities through our experiments.

5.2 Document Classification

To test our morph embeddings in a downstream task application, we employed the same approach to the document classification task using support vector machines. Classification tasks benefit from a better semantic understanding in the constituent word embeddings. Since our morphological segmentation improved the same and also brought down the number of complex inflectional forms by separating the common suffixes, we were able to see a marked improvement of 10% to 15% in both languages, as shown in Table 2.

5.3 Crosslingual Predictions

Finally, we present our crosslingual prediction results in Table 4. Telugu and Kannada belong to the same language family and exhibit similar word order and linguistic structure. However, Hindi is an Indo-European language that exhibits a lot of divergence from these languages. The word ordering is different, and there is no agglutination for Hindi verbs. Each complex Telugu/Kannada word has a group of Hindi words as translation. For example, a Telugu word తయారుచేయాలి (*tayārucēyāli*, "should be prepared") has a four-word sequence (तैयार कर लेना चाहिए, *taiyār kar lēnā cāhi'ē*) in Hindi as its equivalent. To handle this, we averaged over Hindi component embeddings in such cases.

The Kannada to Telugu prediction accuracies improved by almost three times, from 11.36% to 31.8%, when morphs were used instead of words. Similar to the monolingual case, we were able to retrieve several words related to the target translation in the predicted word list. A source word of one POS tag mostly produced words of the same POS tag as translation. Although we have not included any explicit suffix to suffix translations in our training dictionary, suffixes were predicted with remarkable accuracy. Suffix sequences that indicate a part of the source word's meaning were translated into similar sequences in the target language as well. For example, a sequence of

Table 5. Nearest Neighbors from Morph Models (Kan2Tel)

Source Token (Kannada)	Nearest Neighbors (Telugu)	Comments
gāḍha (dark, deep)	*raṅgu* (color), *muduru* (deep/dark), *erupu* (red), *nalla* (black/dark)	Retrieved words could co-occur with the source word in common contexts.
suddi (news)	*vārta* (news), *vilēkaru* (journalist), *sampādaku* (editor), *samācāraṁ* (information), *saṅghaṭana* (incident)	Same as above.
bare (write)	*rāya* (write), *mudrinča* (print), *āmōdimpa* (accept), *prakaṭimpa* (publish), *sarididda* (correct)	A set of verbs related to the source verb *write* was retrieved as the closest words.
vēga (fast)	*vēgaṁ* (fast), *adhikaṁ* (abundant), *śrēṣṭhaṁ* (of high quality), *kramaṁ* (orderly), *dr̥ḍhaṁ* (strong)	Several adjectives indicating positive qualities were retrieved, as the source adjective was used in a positive sense.
śrīlaṅkā (Sri Lanka)	*śrīlaṅka* (Sri Lanka), *mayanmār* (Myanmar), *taivān* (Taiwan), *thāyilāṇḍ* (Thailand), *baṅglādēś* (Bangladesh)	The source word was recognized as the name of a country, and several country names in the target language were retrieved.

suffixes ಲಾಗಿ ದೆ (lāgi de, indicating passive voice, can be roughly translated to "it was done") was predicted to the equivalent Telugu sequence బడిం ది (baḍiṁ di) and its variants బడ్డ ది and బడిన ది (baḍḍa di and baḍina di). These alignments were learned without any explicit rules, and in most of the cases, even the words in the translation dictionary did not contain these suffixes. Some of these interesting nearest neighbors retrieved by the morph model are presented in Table 5.

In the case of Hindi, when the words were predicted using word-level embedding models, the accuracies and similarity scores were quite negligible (approximately two to three words were correctly predicted in the 200-word set). This is expected to be due to the divergence in the respective language structures. As mentioned earlier, morpheme segmentation did not bring down the Hindi vocabulary size by much; there were only a few suffixes (<20) that were segmented to form the morpheme corpus. However, when predictions were tested on Hindi and Telugu morpheme-level models, we observed a positive improvement in the predictions, as indicated in Table 4. Similar to kan2tel translations, the Hindi suffixes also were predicted to their equivalent Telugu suffixes. For example, the Hindi suffix ना (nā) indicating the gerund form a verb was predicted to డం (ḍam), డానికి (ḍāniki), డమే (ḍamē), and డాన్ని (ḍānni), in Telugu, all of which are used with various cases of a gerund in Telugu. Even the similarity scores of the morph models seemed to improve, even if slightly, with morphological analysis.

5.4 Comparison with Fasttext

In the case of fasttext embeddings, we observed that word-level fasttext Spearman rho scores are on par with word-level word2vec, as shown in Table 4. Even when the model failed to predict translations, resulting in inferior top 1 and top 5 scores as presented, we observed that a lot of the predicted candidate translations had the same suffixes and mostly belonged to the same morphological class, even when unrelated by meaning. This was especially true for longer words with complex suffixes. Shorter words showed no such behavior, which can be justified by the fact that fasttext considers a sum of the character n-grams as a feature instead of the word itself. Longer words have more n-gram features, which seems to help with the predictions. This is also the reason behind poor accuracies and similarity scores for morpheme variants of fasttext embeddings. When the words are divided into roots and suffixes, the n-grams from the other divided part of each word are missing when performing the prediction, leading to poor predictions, as opposed to word2vec models.

These results promise that with some refinement to our methods, we can achieve much better translations in low resource language situations. One such refinement could be augmenting the existing word2vec with morpheme features as used in our methodology, along with additional character n-gram features similar to fasttext. Embeddings can also benefit from training by using larger corpora to reduce the sparsity of data, which can lead to better Spearman rho similarity scores in both monolingual and crosslingual cases.

6 RELATED WORK

Bengio et al. [2003] introduced the idea of generating word embeddings as part of the neural network language model. The softmax function required to generate the word embeddings in their model needed extreme amounts of computational power. Their approach was extended in Collobert and Weston [2008] and Collobert et al. [2011] to replace the softmax with a less expensive objective function. Later, the word2vec model, which was even more less computationally intensive than the previous models, was proposed [Mikolov et al. 2013a]. This incited the interest of the NLP research community to explore the applications of word2vec embeddings for several tasks, such as document classification and sentiment analysis.

In a later work, Mikolov et al. [2013c] explored the possibility of using these word2vec embeddings for machine translation by generating and extending lexicons and phrase tables, by making use of a bilingual dictionary. However, their model was not morphology aware.

Luong et al. [2013] proposed to learn morphologically aware word representations to generate more accurate word embeddings for rare words composed of several morphs. Botha and Blunsom [2014] focused upon making use of compositionality of morphemes when building log bilinear models. Qiu et al. [2014] aimed to incorporate morphology into their word representations learned through an RNN. Kim et al. [2016] trained their LSTM base language model at a character level rather than word level and better identified morphological variations.

Similarly, Bojanowski et al. [2016] aimed to evaluate the working of a “bag of character n gram model” upon standard evaluation tasks such as word similarity and analogy. This formed the basis for Facebook’s fasttext word embeddings at a character level.

Cotterell and Schütze [2015] extended the log bilinear model to incorporate morphological information and used a semisupervised approach to learn morphology. Bhatia et al. [2016] treated word embeddings as latent variables that are conditioned upon a prior distribution based upon the morphology, gaining better results on intrinsic word similarity evaluations.

Thus far, crosslingual embeddings have been explored for several different tasks, such as crosslingual sentiment analysis and document classification tasks [Duong et al. 2016].

In their work, Duong et al. [2016] tried to build crosslingual embeddings without a parallel corpus by making use of a huge bilingual dictionary, namely PanLex. Cohn et al. [2017] attempted to construct embeddings for low resource languages by making use of the same large open multilingual dictionary approach. In a later work, Kanayama et al. [2017] explored several different algorithms to generate crosslingual embeddings in addition to the linear transformation method mentioned earlier.

In the context of Indian languages, Ramanathan et al. [2008] proved that reordering English sentences to fit Hindi word order and performing a suffix stripping upon both the languages results in increased BLEU scores. In addition, in the work of Kunchukuttan et al. [2014], it was illustrated that the problem of morphological complexities leads to comparatively lower BLEU scores for Dravidian languages. Despite having low divergence, similar word order, and etymological similarity, the translation among these languages has not been promising. Their work proposed that performing morphological segmentation helps in overcoming the OOV word problem. Some recent work has looked at the word embeddings for low resource languages [Cohn et al. 2017], involving transfer of embeddings from a high resource language to low resource language and identifying verb classes across languages [Vulic et al. 2017].

Several enhancements to the linear projection method of Mikolov et al. [2013c] have been proposed. Upadhyay et al. [2016] discussed some of these approaches and made comparisons among them. Faruqui et al. [2014] used a similar linear transformation approach but used a canonical correlation approach to arrive at the alignments. Xing et al. [2015] proposed normalization of word vectors to improve monolingual similarity. We observed that normalization did not make a noticeable difference to us due to our relatively smaller vocabulary size. Artetxe et al. [2016] generalized some of these ideas to build crosslingual embeddings that preserve monolingual invariance as well. Some approaches, such as those of Luong et al. [2015b] and Tsai and Roth [2016], involved training the embeddings together for all languages involved. Deep learning approaches have also been applied for identifying crosslingual embeddings [Lu et al. 2015]. All of these approaches consider word-level embeddings for making crosslingual predictions. Meaningfully breaking down complex words into a set of simpler morphs for extracting crosslingual similarities is a much less explored idea. Further, these existing approaches work with languages of much higher resources, amounting to corpus sizes of billions of sentences. To the best of our knowledge, we are the first to explore the idea of extracting crosslingual embeddings among low resource, complex, and agglutinative language groups without making use of a high resource language such as English or French.

Seq2seq models employing an encoder-decoder architecture [Sutskever et al. 2014; Cho et al. 2014] have brought about revolutionary changes in NMT in the past 5 years. NMT gracefully handles the variations and subtleties of languages by learning the language characteristics and typical sentence architectures from the training data. NMT, particularly attention-based machine translation [Bahdanau et al. 2014], exhibited comparable results to those of statistical phrase-based machine translation and even outperformed traditional systems for longer sentences [Bahdanau et al. 2014; Luong et al. 2015a]. Since one of the major limiting factors of NMT is the parallel corpus, efforts are being put toward developing an unsupervised variant of NMT [Artetxe et al. 2018; Lample et al. 2018; Fadaee et al. 2017]. Crosslingual embeddings, where embeddings from different languages are projected into a shared space, are making low resource NMT possible [Gu et al. 2018].

7 CONCLUSION

The major motivation of our work was to prove that morphologically aware embeddings are more useful to NLP tasks than those generated at word level. Our approach of performing simple morphological segmentation prior to word embedding generation has benefited both monolingual and crosslingual scenarios, as observed in the results. A full-fledged morphological analysis would benefit the method even more, but we would like to explore it in the future, as it could be complex and needs human language expertise. The bilingual dictionary is basically an alignment strategy. This could be replaced by an attention mechanism in a neural network translation model. However, in our preliminary experiments, straightforward neural net methods proved difficult to converge due to the small size of the dataset. We also wish to augment our corpora in the future and explore even more sophisticated approaches to generate crosslingual embeddings and improve machine translation quality.

Another observation from our experiments was that the character n-grams of fasttext embeddings contributed to similarity scores between longer words with the same suffixes. Therefore, we also wish to explore a method to incorporate this character-level information into our morpheme model.

Our current approach does not explicitly handle *sandhi*s and complex words. We plan to handle these in our future work to build systems with better coverage. As the creation of data and tools is expensive, we believe that low resource languages can benefit from more linguistically aware NLP systems.

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